POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2024
Programme: BE | Full Marks : 100
Course: Applied Physics | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is compound pendulum? Deduce the expression for the time period of compound pendulum and compare this period with simple pendulum. 9

OR

Explain clearly the terms free, damped and forced oscillations. Developed the differential equation of a particle executing damped oscillations. Explain the energy dissipation in such oscillations.

b) A body is propelled through a viscous medium with a speed of 300cm/sec on the propelling force being suddenly removed, it comes to rest after covering a distance of 45m. What is the relaxation time? How long would it be before its speed falls to 30cm/sec? 6

2. a) Define Ultrasonic waves with their properties. Discuss Piezo – Electric Generation of Ultrasonic waves and write down its advantages and disadvantages. 9

b) Entry of people in an auditorium of volume 1.6×10^5 cubic ft and absorption of 10^3 square ft of open window raises the absorption by 600 sq ft. Find the change in reverberation time. 6

3. a) Discuss the principle of propagation of light wave through the Optical Fiber. 9

b) Calculate the refractive indices of the core and cladding materials of a fiber from following data. Numerical aperture (NA) = 0.22 and fractional refractive index change $\Delta = 0.012$. 6

4. a) What is a capacitor? Discuss the discharging of capacitor through resistor. And hence plot the graph between current with respect to time of discharging. 9

b) Derive the Clausius-Mossotti equation with appropriate notations. 6

5. a) Convert the integral form of Maxwell's equations to the differential form. Write the physical significance of each equations. 9

b) Using Maxwell's equation, prove that speed of electromagnetic wave is greater in space then in medium. 6

6. a) State second law of thermodynamics. Discuss working principle of heat engine and heat pump with their coefficient of performance. 9

OR

Discuss the assumptions of Fourier's law of heat transfer. Hence discuss the one dimensional steady state conduction through plane wall.

b) A heat engine operates between a cold reservoir at temperature $T_2 = 300K$ and a hot reservoir at temperature T_1. It takes 200J of heat from hot reservoir and delivers 120J of heat to the cold reservoir in a cycle. What would be the minimum temperature of hot reservoir? 6

7. Write short notes on: **(Any two)** 2×5

a) Inadequacy of classical mechanics

b) Wave function and its significance

c) Second law of thermodynamics with different approaches